Kitchen Planning Skill

Usage Guide and Complete Skill Files

2026-10-09

Documentation Guide

Purpose and audience

This document explains how to use the kitchen-planning skill and includes the complete contents of its three Markdown files, formatted for reading in Word. It is intended for kitchen planners, reviewers, and people integrating the skill into an AI-assisted planning workflow.

The skill supports residential kitchen planning with Germany/EU defaults, millimetres for dimensions, and EUR for costs. It is vendor-neutral and must be adapted to the actual jurisdiction, site, products, and household.

Contents

- Using the skill
- Full skill definition, including metadata
- Full planning reference
- Full source catalog

The headings also support Word's Navigation Pane. Links between the included files lead to the corresponding chapter; external source links remain clickable.

Document status

This is a documentation snapshot, not an executable skill, a product catalog, or a professional installation approval. The Markdown files in the workspace remain the maintained originals. The chapters reproduce their full content; Markdown syntax is rendered as headings, lists, tables, code, and links. The skill's YAML metadata is included explicitly as a code block.

The source catalog is included here for completeness. Its presence in this document does not change the skill's optional-loading policy.

Using the Skill

Installation and invocation

Keep the following structure in the workspace:

```
.github/skills/kitchen-planning/
    SKILL.md
    references/
        planning-reference.md
        source-catalog.md
```

In a compatible VS Code Copilot environment, invoke `/kitchen-planning` or ask for a kitchen-planning task that matches the skill's description. Skill discovery depends on the host's support and configuration. In another AI application, load the skill through that application's instruction/resource mechanism; a Word file alone does not install or activate the skill.

Which files to load

File	Role	Loading policy
SKILL.md	Main workflow, intake, constraints, cabinet assignment, task trials, and deliverables.	Load when the skill is invoked.
references/planning-reference.md	Dimensions with qualifications, 40 check categories, and worked examples.	Read before applying numerical guidance or reviewing installation risks; use its checklist as directed by the skill.
references/source-catalog.md	Ten source entries, URLs, publication details, and research notes.	Optional: load for source verification, provenance/citation questions, or guidance updates, not routine planning.

Local regulations, exact product instructions, and professional checks may still be required for a project. Making the bibliography optional does not make those checks optional.

Example requests

New layout: "Use /kitchen-planning to develop alternatives for my kitchen in Germany. I will provide a measured floor plan, fixed service points, appliance models, household needs, and budget. Identify missing information before claiming the layout is dimensionally checked."

Storage review: "Review this layout for dishwasher unloading. Assign dining cutlery, plates, bowls, cups, and cooking utensils to cabinets. Test the dishwasher and receiving drawers open together, and explain the routes and trade-offs."

Evidence review: "Explain the source and limitations of the suggested preparation width between sink and hob. Consult the source catalog and distinguish ergonomic guidance from installation requirements."

Planning sequence

1. Gather the household brief, inventory, measured geometry, services, and products.
2. Separate hard constraints from recommendations and preferences; preserve unknowns.
3. Arrange food storage, preparation, cooking, cleaning, and clean-storage zones.
4. Assign actual contents before selecting cabinet dimensions and drawer fittings.
5. Compare feasible alternatives using open-state task trials and walkable routes.
6. Deliver the layout, storage schedule, constraint report, and professional-review actions.

Plan for daily activities rather than cabinet symmetry alone. A useful sink-to-hob preparation surface is an ergonomic goal, not a universal EU legal distance. Nearby dish storage is useful only when it remains reachable with the dishwasher and receiving storage open. Cooking tools generally belong at their point of use.

Expected results and limits

The output should explain what goes in each cabinet, how everyday tasks work, which facts are verified, and which constraints remain unresolved. Readiness progresses through `concept`, `dimensionally_checked`, and `ready_for_professional_order_review`. None of these replaces required professional installation approval or separate authorization to place an order.

The skill is written guidance, not a deterministic geometry validator. It does not provide an appliance database, certify legal compliance, or establish hidden services from photographs. Use authoritative geometry/catalog tools where available, and label estimates or missing information honestly.

Maintenance and rebuilding

Edit the maintained Markdown files, then regenerate this document with:

```
python scripts/build-kitchen-documentation.py
```

The builder requires `python-docx`, `pypandoc`, and an available Pandoc executable (for example, provided by `pypandoc_binary`). It reads the latest three files, generates the Word document, and checks that their text content is represented. It does not modify the skill files. Review the generated document before external publication, especially after introducing new Markdown constructs or wide tables.

Full Skill Definition

Source file: `SKILL.md`

```
---
name: kitchen-planning
description: 'Plan or review a residential kitchen: room survey, kitchen layout, sink and hob spacing, dishwashe
r placement, cabinet ordering, drawer organization, storage zones, short walking routes, ergonomics, accessibili
ty, appliance clearances, and installation readiness. Use for kitchen planning, kitchen design, Kuechenplanung,
arranging cabinets, or improving cooking and dishwasher-unloading workflows. Defaults to Germany/EU and millimet
res; adapt rules to the actual jurisdiction.'
---
```

Kitchen Planning

Purpose and Scope

Produce practical, dimensioned residential kitchen proposals with short daily routes, usable storage, and explicit feasibility checks. Plan from activities and contents before choosing cabinet fronts. Here, cabinet "ordering" means arranging cabinets; procurement requires a separate approval gate.

Default to Germany/EU, dimensions in mm, and costs in EUR. Confirm country and, in Germany, Bundesland before applying regulatory requirements. There is no single EU kitchen-clearance code. Commercial kitchens require a separate specialist workflow.

Read the planning constraints reference before applying numerical guidance or reviewing installation risks. It includes rule classifications, qualifications, and short source IDs for traceability. Recheck current local requirements and exact product manuals for each real project.

The source catalog is optional. Load it only to verify a source-backed claim, answer a provenance/citation question, or update the guidance. Do not load it during routine planning merely to resolve source IDs; the operational guidance and its limitations are in this skill and the planning reference. This does not waive project-specific verification of regulations or product instructions.

Non-Negotiable Planning Discipline

- Treat law, applicable technical requirements, product instructions, and verified site constraints as simultaneous constraints. If they conflict, stop and resolve the conflict; a manufacturer cannot waive the law.
- Do not turn a recommendation into a regulation. In particular, do not claim that a sink beside a hob is universally illegal. Usually the problem is missing preparation/landing space; actual installation restrictions are product- and site-specific.
- Unknown is not pass. Missing dimensions, manuals, or service information remain unresolved. Photographs and generated images cannot establish installation readiness.
- Use a planner's authoritative geometry, catalog, and prices where available. Never invent a catalog item, price, service point, clearance, or successful validation result.
- Optimize only after hard constraints pass. A short walking route cannot compensate for a blocked exit, unsafe installation, or inaccessible appliance.
- Offer concepts with explicit assumptions when facts are incomplete. Do not call them order-ready or installation-ready.
- Do not move services, alter structures, purchase items, or overwrite an accepted design without explicit authorization. Draft alternatives should remain reversible.

1. Collect the Brief and Survey

Ask only for missing information that changes the next decision. Start with room dimensions/openings, fixed services, users, appliances, and budget. Collect remaining details progressively.

Input	Required information
Location and scope	Country, region, dwelling type, new build/remodel, ownership/landlord restrictions, required accessibility standard, retained items.
Room	Each wall length, ceiling/soffit heights, angles/diagonals, niches, columns, slopes, and finished floor levels. Measure irregular walls at relevant heights.
Openings	Door/window position, clear opening, frame/trim, sill height, opening direction and swept volume; curtains/blinds and window handles.
Services	Measured water/drain positions and heights, shutoffs, electrical points/circuits, gas, ventilation/exhaust routes, heating, meters, and service access. Mark relocation permission separately.
Structure	Wall construction/fixing capacity, floor/ceiling constraints, existing ventilation, room-air-dependent fireplace or combustion appliance, underfloor services.
People	Household size, number of simultaneous cooks, heights/elbow heights, handedness, reach/lifting ability, mobility aids, children, and future needs.
Activities	Shopping frequency, everyday meals, batch cooking, baking, coffee, entertaining, table setting, cleaning, waste sorting, and current annoyances.
Inventory	Counts and dimensions of dishes, pans, trays, containers, food, appliances, and waste bins; frequency and typical weight of use.
Products	Exact models or provisional envelopes, installation manuals, cabinet system, appliance niches, front/hinge system, handle projections, and worktop material.
Commercial	Total installed budget, service relocation allowance, delivery/access constraints, timing, and priorities ranked by the user.

For each critical fact record `value`, `unit`, `source`, `verification_state`, and measurement uncertainty where known. Use `unknown`, `inferred`, `user_verified`, or `professionally_verified`. Preserve raw measurements and identify the finished reference surfaces.

If only a photo is supplied, request a measured plan or verified reference dimensions. Hidden pipe routes, wall strength, electrical capacity, and regulatory compliance cannot be inferred from appearance.

2. Build the Constraint Register

Give each rule a stable ID and classify it:

Class	Meaning	Handling
`regulatory`	Verified applicable legal/code requirement, with jurisdiction and edition.	Hard; no user override.
`manufacturer`	Exact installation/compatibility requirement for the selected product.	Hard; change the product or design if necessary.
`site`	Verified room, structural, service, or geometric limitation.	Hard unless an approved change creates a new verified condition.
`user_must`	Explicit user requirement, including a fixed budget if stated.	Hard for this brief; ask before relaxing.

Class	Meaning	Handling
`recommendation`	Sourced ergonomic/design recommendation or labeled planning heuristic.	Soft; document the consequence of departures.
`preference`	Style, symmetry, convenience, or other discretionary choice.	Soft; trade off transparently.

Each record needs `id`, `class`, `applicability`, `requirement`, `measurement_definition`, `source`, `status`, `affected_items`, and `remedy_or_next_action`. Status is `pass`, `fail`, `unknown`, or `not_applicable`; explain exclusions. Preserve recommendation deviations rather than silently deleting them.

Complete every category in the reference's installation checklist. Record applicability separately from compliance: an unconfirmed rule is not automatically irrelevant.

3. Arrange Functional Zones

Use five zones as an adjacency model, not a compulsory left-to-right sequence [S1, S2]:

1. Food storage: refrigerator/freezer and dry provisions.
2. Preparation: the principal clear work surface and preparation tools.
3. Cooking: hob, oven, heat-safe landing surfaces, and cookware.
4. Cleaning: sink, dishwasher, waste sorting, and cleaning supplies.
5. Clean storage/serving: dishes, dining cutlery, glasses, and serving items.

Trace these distinct workflows:

- Shopping arrival -> grocery landing -> refrigerator/pantry.
- Food storage -> sink when needed -> preparation -> cooking -> serving.
- Table return -> scrape/sort -> sink as needed -> dishwasher. Do not prescribe routine pre-rinsing; follow appliance guidance.
- Dishwasher -> clean storage -> table/serving.
- Hob -> sink for draining hot cookware, without avoidable traffic crossings.
- Coffee/breakfast and snack access without walking through active cooking.

Keep through-traffic outside the principal work area. The refrigerator-sink-hob triangle is a diagnostic, not a mandatory geometry or a substitute for these workflows. A one-wall kitchen can work without a triangle. Mixed-handed or multi-cook households need task trials, not automatic mirroring.

Sink, Preparation Surface, and Hob

Prefer `sink -> continuous preparation surface -> hob`, or the mirror image. Blum recommends approximately 900 mm of preparation width, with 800 mm as its lower ergonomic recommendation [S3]. These are not German legal minimums. More space may be needed for two cooks or extensive baking.

Measure usable clear surface between fixture edges, excluding an occupied drainer, permanent appliances, or other obstructions. Record actual usable depth too. Separately verify cut-out-to-cut-out webs, edge distances, and supports from the worktop fabricator; a visible surface gap does not prove structural suitability.

Avoid placing a tall cabinet across this continuous workflow. Keep the preparation surface genuinely available after locating the kettle, coffee machine, knife storage, and drying area. Where the room cannot accommodate the recommendation, disclose the compromise and test an adjacent alternative preparation surface.

Dishwasher and Clean Storage

- Start with the dishwasher beside the sink, subject to plumbing, appliance instructions, and usable access. Prefer a side outside the main preparation stance when feasible.
- Compare both sides of the sink. Do not prescribe left/right solely from handedness.
- Locate everyday dining cutlery, plates, and bowls in an adjacent or nearby accessible drawer bank. Locate daily glasses and cups within comfortable reach nearby [S4].
- Aim for a comfortable turn or short step during unloading. This is a qualitative target, not a universal distance limit.
- Check dishwasher fully open, racks extended, destination drawer open, and a person standing in the unloading position. Neither nominal adjacency nor cabinet-centre distance establishes usability.
- Across an aisle can work only if the actual door/drawer projections, standing space, reach, and circulation work together. Never require stepping over a dishwasher door.
- At an inside corner, check perpendicular drawer collisions and access to the sink/waste. Move the dishwasher or storage bank if fillers alone cannot solve the conflict.
- Keep dining/table access convenient without routing other people through the hot zone. Do not put every item beside the dishwasher: cooking tools still belong at their principal point of use.
- Consider a raised dishwasher for users who benefit, but validate approved support, niche, front mechanism, rack heights, and adjacent unloading space.

4. Assign Contents Before Cabinet Sizes

Create the storage schedule first. Then fit catalog modules around the fixed zones.

Contents	Preferred home	Checks
Dining cutlery	Shallow upper drawer near dishwasher and dish storage.	Organizer fits internally; accessible with dishwasher open.
Everyday plates/bowls	Full-extension drawer in the same or nearby bank.	Plate diameter, stack height, restraint, loaded runner capacity, comfortable lifting height.
Glasses/cups	Reachable nearby shelf, lift-front cabinet, or suitable drawer.	Safe reach, glass height, retention, opening clearance; not forced overhead.
Preparation knives/tools	Upper drawer at principal preparation surface.	Blade protection, child security, separation from dining cutlery.
Boards/trays	Vertical divider storage near preparation/oven.	Actual tray dimensions and removal path.
Pans/pots/lids	Full-extension storage beside or below hob where permitted.	Heat separation, hob underside, ventilation and extractor/oven volume.
Spices/oils	Near preparation/cooking, protected from heat/light/moisture.	Not directly exposed to hob heat or steam.
Mixing bowls, scales, baking tools	Preparation/baking zone.	Nesting, appliance weight, lifting distance, and usable drawer height.
Dry food	Pantry/pull-outs near grocery landing and preparation.	Stock volume, visibility, shelf adjustment, runner loads.
Waste/recycling	Accessible from preparation and sink.	Local sorting streams, bin removal, odours, and access while dishwasher is open.
Cleaning chemicals	Separate secure storage, often near sink.	No food/clean-utensil mixing; child access, plumbing and leak clearance.
Food containers/lids	Near preparation, leftovers packing, and dishwasher.	Paired or divided organization; capacity for actual inventory.

Contents	Preferred home	Checks
Small appliances	At use location or reachable storage.	Operating footprint, lid opening, steam, power, ventilation, weight.
Rarely used items	Less accessible but safe locations.	Do not put heavy objects overhead simply because they are infrequent.

Prefer full-extension drawers for frequently used base storage [S4, S5]. Upper shallow drawers suit small tools; deeper drawers suit bulky items, but choose heights from actual contents, not a fixed three-drawer formula. Prioritize heavy frequently used items at a comfortable lifting height rather than automatically on the lowest level.

Compare wide drawer banks with multiple narrow cabinets using usable capacity, load, cost, and multi-user access. A wide open drawer can block another cook. Compare corner mechanisms with a deliberately unused corner plus accessible straight drawers; maximum gross volume is not necessarily best value.

For each cabinet record: `id`, `zone`, `catalog_id_or_provisional`, external dimensions, usable internal dimensions, contents, load, opening direction/extension, fittings, and reason for location. Include fillers, end panels, scribes, plinths, supports, and tolerances in each run. Never assume nominal width equals usable storage or appliance niche width.

5. Generate and Check Alternatives

Develop two or three genuinely different feasible options when geometry permits, for example retained services versus improved workflow. Do not force an island or a U-layout into a room that cannot support it.

For each option:

1. Place immovable constraints and reserve circulation/accessibility envelopes.
2. Locate sink, principal preparation area, hob/extraction, and refrigerator landing.
3. Fit dishwasher, waste, and nearby clean storage as a working group.
4. Allocate remaining storage and secondary tasks from the inventory.
5. Fit actual modules, fillers, worktops, panels, and service voids.
6. Check 3D fit, closed projections, full swept openings, service access, ventilation, and installation tolerances.
7. Run the task trials below. Repair hard failures before comparing soft preferences.

Required trials: unload a full dishwasher; prepare/cook a typical meal; drain a heavy pot; set/clear the table; put away groceries; remove waste; and let a second person reach drinks while cooking continues. Include the actual accessibility/child-use scenarios in the brief.

Check necessary simultaneous openings, especially dishwasher plus receiving drawer, refrigerator door plus internal drawer, sink/waste plus dishwasher, and perpendicular corner units. Not every door must open simultaneously, but the normal task combinations must work. Record remaining conflicts and their consequences.

Measure travel along collision-free walkable paths between usable standing positions, with task-specific openings represented. Do not measure straight through an island or across an open door. Without a geometry/path tool, use a clearly labeled manual estimate or `unknown`; do not fabricate precision.

Compare Everyday Effort

Use lexicographic priorities: hard feasibility first, essential usability/storage second, daily effort third, then cost/style trade-offs within the agreed brief.

For a transparent optional distance estimate, calculate:

```
daily_travel_m = sum(task_cycles_per_day * sum(trip_count_per_leg * walkable_leg_length_m))
```

Define each leg as a one-way movement. Include return legs explicitly where they occur. For unloading, use realistic batches carried per trip, not one trip per plate and not an implausibly heavy stack. Use the same inventory, carry capacity, and task frequencies for every option. Label assumed frequencies and distances.

Report bending/high reaching, awkward lifts, door conflicts, and traffic crossings separately; do not add metres, counts, and money without explicit normalization and user-agreed weights. Never label a heuristic score a certified optimum or claim percentage savings without a baseline calculation.

6. Deliver the Plan

Return a concise recommendation followed by:

1. Brief, assumptions, missing measurements, and readiness status.
2. Dimensioned layout, orientation, appliance locations, and clear work surfaces. If no drawing tool exists, supply wall-by-wall cabinet sequences with dimensions, not an invented rendered plan.
3. Cabinet/drawer schedule specifying exactly what goes where.
4. Dishwasher-unloading explanation: standing position, receiving drawers/cup storage, openings, paths, and compromises.
5. Alternatives table: hard failures/unknowns, preparation space, usable storage, travel estimate, ergonomics, installed cost confidence, and trade-offs.
6. Constraint report with rule IDs, measurement definitions, evidence, pass/fail/unknown, and remedies.
7. Action list assigning survey, electrical, plumbing, ventilation, structural, accessibility, and installation checks to the appropriate reviewer.

Use the status `concept`, `dimensionally_checked`, or `ready_for_professional_order_review`. The last status requires verified survey, exact product compatibility, complete parts/cost scope, and no unresolved hard constraint. It is not professional installation approval. When required reviewers approve, record who approved what and against which plan/manual revisions before any separate order authorization.

If no feasible layout exists, identify the conflicting constraints and smallest changes to consider, such as reducing an island, changing an appliance, or obtaining approval to relocate services. Never silently shrink a required clearance or alter a user must-have.

Quality Checks for This Skill

Before delivering a plan, verify these counterexamples are handled correctly:

- Sink beside hob: report preparation-space and installation checks, not an invented EU prohibition.
- Dish drawer close to dishwasher but behind its open door: reject the unloading arrangement despite the short nominal distance.
- Short route through an island: replace it with a walkable route or mark distance unknown.
- Attractive layout with unknown hood clearance: keep installation readiness blocked.
- Single-wall or mixed-handed kitchen: use task-based zoning without forcing a triangle or handedness rule.
- Wheelchair brief: invoke applicable accessibility checks and user trials; a generic aisle recommendation is insufficient.
- Best geometric score but insufficient plate storage or overloaded drawers: do not recommend until capacity/load are resolved.

Full Planning Reference

Source file: `references/planning-reference.md`

Kitchen Planning Evidence and Constraints

Research checked: 2026-10-09. Scope: residential kitchens, Germany/EU default. This is an original synthesis of public guidance, not a reproduction of a standard, a legal opinion, or an installation specification.

Evidence Boundaries

AMK, Blum, and IKEA offer useful planning guidance; their editorial advice is not law. Product marketing is not proof of compatibility or universal ergonomic superiority. A named standard is mandatory only where applicable through law, contract, or the project brief; establish that applicability with qualified reviewers.

Short source IDs refer to the optional source catalog, which holds URLs, publication details, and research notes. Do not load it for routine planning; consult it only for source verification, provenance/citation questions, or guidance updates. Rule classifications and operational qualifications remain in this reference.

The full DIN standard and project-specific appliance installation manuals were not examined. Numerical clearance requirements for a real product therefore remain to be supplied. Verify current applicable requirements and exact product instructions for each project; the research date does not establish a document's technical currency.

Some general advice is too absolute to reuse literally. For example, induction surfaces can remain hot from cookware, lift-up doors still need head/ceiling/reach checks, and handedness does not determine a whole household's best layout. Apply task and product checks rather than repeating marketing assurances.

Dimensions and How to Use Them

All dimensions below are in mm. Label the rule class and source whenever presenting them to a customer.

Topic	Starting guidance	Status and measurement
Main preparation width	Approximately 900; Blum advises at least 800 for comfortable preparation.	Published ergonomic recommendation [S3], not a statutory minimum. Clear usable work surface, preferably between sink and hob; not centre-to-centre distance.
Single-cook working aisle	Explore roughly 1000-1200 initially.	This skill's concept-sizing heuristic, not a sourced legal minimum. Measure the narrowest gap between finished closed projections, then test open appliances and the user.
Shared working aisle	Explore roughly 1200-1500 initially.	Concept-sizing heuristic only. Simultaneous work, carry tasks, and passage can need more; extra width also lengthens transfers.
Standing preparation height	Start near measured elbow height minus 150.	AMK recommendation [S1]; test in usual footwear with a mock-up. Sink depth and cookware change effective task height. No universal 900-high worktop.
Accessible kitchen movement space	The DIN 18040-2 secondary summary describes 1200 x 1200, and 1500 x 1500 for wheelchair use; front clearance depths of 1200 and 1500 respectively.	[S7] is a secondary explanation, not the standard itself. Confirm current applicable provisions, including the wheelchair-specific category, and all other access requirements. These are not universal minimum kitchen dimensions.
Dishwasher to dish storage	Comfortable turn or short step where feasible.	Adjacency principle [S4]; no universal maximum distance established by this research. Open-state usability determines success.

Topic	Starting guidance	Status and measurement
Hob/oven/fridge landing	Reserve usable nearby surface sized for the actual pan, tray, or grocery load.	Functional planning recommendation. Do not invent a universal legal landing width. The hot-tray route must be safe with the appliance door open.
Hood, hob, combustible surfaces	Use exact product and applicable local requirements.	No generic numeric default. Check all applicable instructions; resolving a conflict may require changing products.
Cut-out webs, overhangs, supports	Use material/fabricator specifications.	No generic numeric default; material, thickness, cut-outs, support and installation details matter.
Drawers, fronts, fillers, service voids	Use selected cabinet/appliance system drawings.	Catalog-specific; never insert an arbitrary filler and assume collision/installation issues are resolved.

A concept aisle width is not the free width remaining during use. For an illustrative opposing arrangement only, a 1200 aisle minus a 600 dishwasher-door projection minus a 500 drawer projection leaves just 100 between the open fronts. That is not adequate standing space between them. Real doors, racks, handles, body envelopes, and sideways access require actual drawings and task simulation. The numbers here are hypothetical, not product specifications.

Installation and Usability Checklist

Use the IDs below in the project's constraint register. These are check categories: obtain evidence to establish the exact hard requirement or recommendation for each project. Mark unknowns explicitly.

ID	Check	Evidence or required outcome
GEO-01	Verified geometry	Finished dimensions, out-of-square walls, slopes, trims, levels, and measurement tolerance are represented. No fit claim based only on photos.
GEO-02	Circulation and exit	Doors, clear passages, room access and applicable escape requirements remain usable. Assess appliance-open task states and occupants.
GEO-03	Collision volumes	Check full door/drawer travel, handles, lift fronts, internal pull-outs, adjacent fronts, and head clearance. Closed-box collision tests are insufficient.
GEO-04	Windows and heat	Tap does not prevent required window operation. Window/blind/curtain position, hob heat, drafts, splash exposure and required access are reviewed.
GEO-05	Seating and island	Account for occupied seats, pulled-out chairs, knee space, passage and hot/splash zones. Do not double-count the same area for incompatible simultaneous uses.
FIT-01	Cabinet run and niches	Include end panels, corner solutions, fillers, scribes, wall tolerances and actual appliance niches, not just nominal cabinet widths.
FIT-02	Front and plinth systems	Integrated dishwasher front weight/height, fixed or sliding hinge, plinth sweep, mounting and countertop protection match the exact model.
FIT-03	Refrigerator operation	Required opening angle permits internal drawers/shelves to operate and be removed; wall stand-off, hinges and ventilation are correct.
FIT-04	Hob underside and extraction	Body, induction cooling, required separation, cables, downdraft motor/duct/filter and drawers do not compete for the same volume.
FIT-05	Oven and microwave	Approved niche, ventilation, door travel and shelf reach; hot vessels can be removed safely to a landing surface. A high oven must not require lifting hot trays above the user's safe reach.
FIT-06	Raised appliances	Support structure, fixing, manufacturer approval and reachable controls/racks are verified. Do not put a floor-supported appliance on an improvised shelf.
WAT-01	Sink assembly	Bowl, drainer, clips, tap, waste, trap, hoses, drawers and bins fit together; servicing remains possible.

ID	Check	Evidence or required outcome
WAT-02	Water and drainage	Qualified installer verifies supplies, pressure, hot-water compatibility, drainage route/levels, appliance hose limits and required backflow protection. Proximity alone proves none of these.
WAT-03	Shutoff and leak access	Isolation, connections and leak-protection equipment remain accessible as required; no pinched hoses or inaccessible maintenance points.
WAT-04	Special water equipment	Boiling/chilled/carbonated/filter systems have approved taps, tank/filter/cylinder space, ventilation, power, maintenance and child protection.
ELE-01	Electrical design	Qualified electrician verifies available supply, connected loads, circuits, protective devices, earthing, isolation and socket installation under applicable rules.
ELE-02	Socket and cable placement	Plan for all fixed and small appliances, lighting and powered fittings. Keep cords out of wet/hot areas and across no circulation route; verify plugs do not consume a prohibited niche/service volume.
ELE-03	No invented wet-area rule	Do not transpose bathroom electrical zones into a kitchen or assert a blanket socket-to-sink distance without an applicable source. Obtain the electrician's project-specific layout.
GAS-01	Gas equipment	Approved professional checks supply, isolation, installation, ventilation and appliance-specific combustible clearances. No DIY connection instructions.
VEN-01	Extraction choice	Establish exhaust or recirculation early. Check hood capture at the actual hob, installed airflow, noise, filter access, duct dimensions, bends, outlet and replacement-air strategy.
VEN-02	Recirculation limits	Recirculation normally does not remove cooking moisture outdoors and is not a substitute for necessary room ventilation. Check system-specific requirements.
VEN-03	Combustion interaction	Exhaust plus a room-air-dependent fireplace/heater triggers professional pressure/backdraft review with the responsible chimney sweep and ventilation specialist. Do not assume a window switch alone makes the system safe.
VEN-04	Building permissions	Check landlord/owners' association approvals, facade penetrations, fire separations, communal shafts and whole-dwelling ventilation interactions.
TOP-01	Worktop structure	Fabricator verifies cut-out spacing, edge webs, joint locations, supports, spans, overhangs, thermal limits and installation method.
TOP-02	Water and steam protection	Seal/protect cut-outs, joints, sink area and dishwasher edges as specified; account for automatic-opening dishwasher steam.
STR-01	Fixings and stability	Suitable wall fixings, rails, suspended loads, tall-unit restraint, island support and loaded drawers. Check floor/underfloor services before drilling.
STR-02	Structural alterations	No wall removal or structural drilling assumption without qualified assessment and required permissions.
USE-01	Preparation and landings	Real unobstructed preparation and grocery/hot-vessel landing space remains after permanent countertop items are placed.
USE-02	Unloading	Dishwasher, rack, receiving storage and user can occupy needed positions together; no stepping over the door or excessive reaching.
USE-03	Waste and hygiene	Scrape/sort/load route works; bins can be removed and cleaned. Separate chemicals from food/clean items; provide cleanable food-contact surfaces and drying space.
USE-04	Storage capacity and loads	Actual inventory fits internally with chosen organizers and runner ratings. Record spare capacity if requested; do not invent a mandatory percentage.
USE-05	Multi-user tasks	Test concurrent cooking, dish handling, drinks access and table setting; identify crossings and bottlenecks.

ID	Check	Evidence or required outcome
ACC-01	Accessibility	Confirm required standard/category; check turning and approach spaces, knee/toe clearance, seated reach, controls, grip/force, task height and protected hot pipes. Test with the actual user/mobility aid.
SAFE-01	Children and burns	Secure knives/chemicals, review reachable controls and hot-water dispensing, prevent pot-handle exposure to traffic/seating. Induction still has residual heat; child locks do not replace supervision.
SAFE-02	Flooring and edges	Review slip/trip risk, thresholds, protrusions, sharp corners and cleanability. Do not let loose mats create an access hazard.
LGT-01	Lighting	Illuminate preparation, sink and hob without the user casting a working shadow. Check glare, contrast, controls and usable light beneath cabinets; lighting drivers remain serviceable.
ENV-01	Comfort and maintenance	Compare actual appliance noise, energy/water data, extractor filter costs, cleanability, material heat/moisture resistance and repairability. Verify current product labels.
SRV-01	Replacement access	Appliances can be isolated, removed and replaced after flooring/worktops are installed; filters, traps, drivers and fixings can be serviced.
ORD-01	Complete order scope	Include appliances, cabinets, fronts, hinges, runners, handles, fillers, panels, plinths, feet, rails, supports, worktops, cut-outs, lighting, extraction, fittings, delivery, installation and service work.
ORD-02	Release gate	Final survey, product/manual revisions, costs, permissions and responsible professional checks match the exact proposed order. Obtain explicit order authorization.

Worked Arrangement: Pattern, Not Template

An illustrative working run can be:

```
[dish/cutlery drawers] [dishwasher] [sink] [prep drawers + clear worktop] [hob/cookware]
```

Daily cups/glasses can occupy comfortably reachable storage above the dish bank or another nearby accessible location. Waste can be under the sink or in a separate nearby pull-out if loading, preparation and bin removal all work. Do not consume the only preparation drawer space without a replacement.

This pattern places clean storage on one side of cleanup and preparation on the other. It is neither a complete kitchen nor a mandatory sequence: refrigerator/pantry, oven, landing surfaces, end conditions and services must still be located. Mirror or split the run when the room and users benefit. A hob at the run's exposed end still needs its verified clearances and usable landing arrangement.

For an L-layout, a drawer around the corner from the dishwasher may be physically close but impossible to open during unloading. For a galley, opposing dish drawers may work or may block the user's standing position. The decisive check is the open-state task, not the diagram's apparent neatness.

Transparent Travel Example

Suppose two otherwise feasible options each need four storage transfers per dishwasher cycle. Assume three return legs to the dishwasher before the final transfer, one cycle daily, and equal one-way path lengths in either direction:

- Option A: 1.0 m per leg -> `(4 + 3) * 1.0 = 7.0 m` per cycle.
- Option B: 2.5 m per leg -> `(4 + 3) * 2.5 = 17.5 m` per cycle.
- Difference: 10.5 m per cycle under these assumptions, excluding any final return and other kitchen tasks.

These are illustrative distances and batch counts, not research results. Do not extrapolate time or lifetime savings without measurements. A comfortable turn may produce almost no walking but still involve awkward reaching; report that separately. An option with an obstructed unloading path is infeasible regardless of its calculated distance.

Maintenance and Unverified Requirements

The checklist is deliberately broad but not exhaustive. Actual regulatory requirements, permitted clearances, and compatibility depend on the site, products and users. Obtain the current applicable primary sources, exact installation manuals and qualified reviews before converting any unresolved checklist item into a pass.

Revisit this reference when products, material systems, regulations or source guidance change. Keep the general workflow vendor-neutral; replace examples with the project's authorized catalog without losing the open-door, capacity, accessibility and safety checks.

Full Source Catalog

Source file: `references/source-catalog.md`

Kitchen Planning Source Catalog

Optional research reference. Load only to verify a source-backed claim, answer a provenance/citation question, or update the guidance. Routine planning does not require this file; the rules, classifications, and operational qualifications are in the skill and planning reference.

Research checked: 2026-10-09. Scope: residential kitchens, Germany/EU default. The sources below were retrieved for the stated topics. The full DIN standard, paid NKBA handbook, and project-specific appliance installation manuals were not examined. The retrieval date does not establish technical currency; verify current applicable requirements and exact product instructions for each real project.

Sources

S1. AMK: Planen und gestalten

https://www.amk-ratgeber-kueche.de/kueche/planen-gestalten

German kitchen-industry association guidance. Supports functional zones, point-of-use storage, approximately 150 mm elbow-to-worktop difference, individualized working heights, full-extension drawers, early service planning, lighting and accessibility considerations. This is consumer guidance, not a regulation or a complete technical standard. Do not adopt every safety/marketing statement literally; see the evidence boundaries in the planning reference.

S2. IKEA Germany: Gestaltungsideen fuer reibungslose Ablaeufe

https://www.ikea.com/de/de/rooms/kitchen/how-to/kueche-effizient-planen-pub6d21e890/

Supports generous preparation space between sink and hob, proximity of waste and dishwasher, and placing cookware/tools at their point of use. Uses the work triangle as a general planning aid. Does not establish a statutory sink-to-hob distance.

S3. Blum: Speisenvorbereitung leicht gemacht

https://www.blum-inspirations.com/de-eu/wissen/speisenvorbereitung-leicht-gemacht-der-passende-schrank-unterstuetzt-sie-dabei

Manufacturer ergonomic guidance, article dated March 2022. Recommends a preparation area between sink and hob, approximately 900 mm wide, and advises at least 800 mm; preparation tools and bowls belong nearby in organized drawers. These dimensions are ergonomic recommendations, not proof of product clearance or German code compliance.

S4. Blum: Geschirr perfekt organisiert

https://www.blum-inspirations.com/de-eu/wissen/geschirr-perfekt-organisiert-im-unterschrank-mit-auszuegen

Manufacturer guidance, article dated June 2022. Supports dining cutlery and dishes together in full-extension base storage, nearby cups/glasses, and proximity to the dishwasher to reduce unloading travel. It does not supply a universal maximum distance or guarantee collision-free access.

S5. Blum: Die ergonomische Kueche

https://www.blum-inspirations.com/de-eu/wissen/die-ergonomische-kueche-das-muss-man-beachten

Manufacturer guidance, article dated July 2021. Explains access benefits of full-extension base drawers and overhead lift fronts versus projecting doors. Benefits still depend on actual reach, opening height, operating force and room geometry; powered fittings are optional, not necessary for every accessible kitchen.

S6. AMK: Geraetetechnik

https://www.amk-ratgeber-kueche.de/kueche/geraetetechnik

Supports early extraction planning, the importance of capture efficiency, storage consumed by downdraft equipment, and appliance feature/energy/noise considerations. Room-volume airflow rules of thumb alone do not constitute a ventilation design. The combustion-safety checklist in the planning reference is a professional-review trigger, not a quoted clearance rule from this page.

S7. nullbarriere.de: DIN 18040-2 Kuechen, Essplatz

https://nullbarriere.de/din18040-2-kuechen.htm

Specialist secondary summary of German dwelling accessibility guidance. Describes 1200/1500 movement-space dimensions and wheelchair considerations. It is not the official DIN text. Verify the current standard, project category, legal/contractual applicability and complete requirements before claiming conformity.

S8. IKEA Germany: Kuechenplanung

https://www.ikea.com/de/de/rooms/kitchen/kuechenplanung-pub6003162f/

Supports surveying in mm, marking openings/services, and staged planning, checking, ordering and installation. Links to current system-specific planning and installation documents. Those documents and the chosen product manuals must be obtained for a real order.

S9. IKEA Germany: Schubladen organisieren

https://www.ikea.com/de/de/rooms/kitchen/how-to/kuechenschraenke-und-schubladen-organisieren-8-ideen-die-zeit-und-platz-sparen-pube8ed7a50/

Supports full-extension storage, functional grouping, protected knives, organizers and reachable everyday contents. Named fittings are examples, not a recommendation to adopt one vendor.

S10. NKBA Planning Guidelines: Regional Boundary

https://nkba.org/planning-guidelines/

The public overview describes guidance intended for US/Canadian design practice and its code context. Only the overview was consulted, not the paid handbook. Do not label NKBA dimensions as German/EU requirements or cite handbook clauses not actually verified.